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Short communication

Automatic construction of an anatomical coordinate system for three-dimensional bone models of the lower extremities – Pelvis, femur, and tibia

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ABSTRACT

Automated methods for constructing patient-specific anatomical coordinate systems (ACSs) for the pelvis, femur and tibia were developed based on the bony geometry of each, derived from computed tomography (CT). The methods used principal axes of inertia, principal component analysis (PCA), cross-sectional area, and spherical and ellipsoidal surface fitting to eliminate the influence of rater's bias on reference landmark selection. Automatic ACSs for the pelvis, femur, and tibia were successfully constructed on each 3D bone model using the developed algorithm. All constructions were performed within 30 s; furthermore, between- and within- rater errors were zero for a given CT-based 3D bone model, owing to the automated nature of the algorithm. ACSs recommended by the International Society of Biomechanics (ISB) were compared with the automatically constructed ACS, to evaluate the potential differences caused by the selection of the coordinate system. The pelvis ACSs constructed using the ISB-recommended system were tilted significantly more anteriorly than those constructed automatically (range, 9.6–18.8°). There were no significant differences between the two methods for the femur. For the tibia, significant differences were found in the direction of the anteroposterior axis; the anteroposterior axes identified by ISB were more external than those in the automatic ACS (range, 17.5–25.0°).

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1. Introduction

The bony coordinate system reported by Grood, Suntay (1983) has the advantage of being easily described to report the kinematic data of a joint, and has been used as a set of widely acceptable standards. Although various recommendations for the bony coordinate system have been published in different studies (Yoshioka et al., 1987, 1989; Bergmann et al., 2001; Tashman, Anderst 2003; Sato et al., 2004; Kang et al., 2005; Miki et al., 2007; Enomoto et al., 2013), there is a lack of standardization of the bony landmarks used in its construction. This makes comparisons among various studies difficult and potentially leads to errors in the results. Moreover, although the bony coordinate system is used as a criterion for all evaluations, most constructions are determined by manual digitization of bony landmarks. Because landmarks are not points but relatively large and curved areas, their identification by manual digitization is subjected to rater variability (Della Croce et al., 1997; Della Croce et al., 2005; Della Croce et al., 1999). This leads to low between- and within- rater repeatability, especially at deformed joints. Therefore, reliable methods of determining an anatomical coordinate system (ACS) are required.

In this study, we developed automated methods of constructing patient-specific ACSs for the pelvis, femur, and tibia based on the bony geometry of each, derived from computed tomography (CT). We then compared ACSs recommended by the International Society of Biomechanics (ISB) with the automatically constructed ACS, to evaluate potential differences caused by selection of the coordinate system.

2. Methods

Approval from the Institutional Review Board (IRB) of Niigata medical center was obtained before undertaking this study.

2.1. 3D bone models

Five volunteers (two male, three female; aged 65.0 ± 11.2 years) with no history of joint injury were randomly chosen. CT images of lower extremities were

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acquired ($0.76 \times 0.76 \times 2.0$ mm^3; Asteion, Toshiba Corp., Tokyo, Japan). The protocols were 120 kVp and current-time product ranged from 105 to 130 mAs. CT-based 3D bone models of the pelvis, right femur, and right tibia including fibula were then generated using the 3D reconstructing software ZedView (LEXI Co. Ltd., Tokyo, Japan). Although each 3D bone model has a CT based coordinate system, the position of the bone while performing CT scans influenced the direction of the CT-based coordinate system; therefore, a reliable temporary coordinate system was mathematically constructed first.

2.2. Pelvic ACS

We introduced the anterior pelvic plane (APP) described by Lewinnek et al. (1978) to construct the pelvic ACS. The pelvis model was considered as a rigid body; its three principal axes of inertia (eigenvectors of the tensor of inertia) were used as a temporary coordinate system. Since the surface of 3D bone models was represented by numerous triangular facets generated from CT images, area of the facet was considered as imaginary mass, and centroid of the facet was used for calculating the moment arm around the center of mass (Gonzalez-Ochoa et al., 1998).

Based on the temporary coordinate system, vertices were automatically divided into four regions. The dividing line between left and right regions was defined as the midline in the coronal plane. Another dividing line was defined as the midline between the height of maximum width and the height of the most distal point, to determine the proximal–distal region. The four most anterior points in the four regions were automatically calculated to determine APP. Next, two points in the proximal regions were defined as the two ASIS; the midpoint between the two points in the distal regions was defined as PS. ACS for the pelvic model was constructed using these landmarks (Fig. 1).

2.3. Femoral ACS

The center of the femoral head and the geometric center axis (GCA) (Walker et al., 1985; Asano et al., 2001; Eckhoff et al., 2001) were introduced to construct the femoral ACS. A temporary coordinate system was constructed using principal component analysis (Pearson, 1901). Since the femur is categorized as a long bone, three principal component vectors described three anatomical directions (Fig. 2).

Based on the temporary coordinate system, cross-sectional images of the coronal plane were created at intervals of one millimeter, from the most posterior point to the deepest point of the intercondylar fossa. Contour tracking on each image was performed to obtain the outlines of two posterior condyles. The same procedure was performed for the femoral head using the cross-sectional images of the transverse plane. Only the medial contours toward the most proximal point were extracted as the outline of the femoral head. Next, a spherical approximation of each contour using the least squares method was performed; centers of the three spheres were used to construct the anatomical coordinate system of the femoral model (Fig. 2).

2.4. Tibial ACS

The temporary coordinate system was constructed as the femoral ACS. The first principal component vector was used as the *Z*-axis (Fig. 3). Since the first principal component vector that mathematically described the longitudinal axis is mostly not influenced by the direction of fibula, and the fibula also has some important landmarks such as head of the fibula or tip of the lateral malleolus, the fibula was included in the PCA determination. Moreover, tibia and fibula are not separated easily during segmentation, in order to reduce time, the fibula is not routinely separated from tibia in our model construction. The shape of the tibial plateau was used to define the direction of the transverse plane. Contour tracking was performed on cross-sectional images, and the largest outline was calculated. Next, after approximating the outline into an ellipse, the major and minor axes were defined as the *X* and *Y*-axes (Fig. 3). The origin was defined as the center of an ellipse on the *X*–*Y* planes and translated to the most proximal level.

2.5. Evaluation

After each automatic ACS was constructed on each 3D bone model using the described algorithm, ACSs recommended by the ISB were determined by manual digitization (Wu et al., 2002) (see detailed descriptions in the Appendix). Manual digitization was performed by five randomly chosen raters on each 3D bone model. Differences in rotational components between the automatic ACS and ISB-recommended ACSs were compared. Statistical analysis was performed using the intra-class correlation coefficient (ICC) to evaluate the agreement between the five raters.

3. Results

Automatic ACSs were successfully constructed on each 3D bone model using the algorithm described. All constructions were performed within 30 s; furthermore, between- and within- rater errors were zero, owing to the automated nature of the algorithm.

Some notable differences in rotational components between the automatic ACS and ISB-recommended ACSs were observed. Pelvises with ISB-recommended ACS were tilted significantly more anteriorly

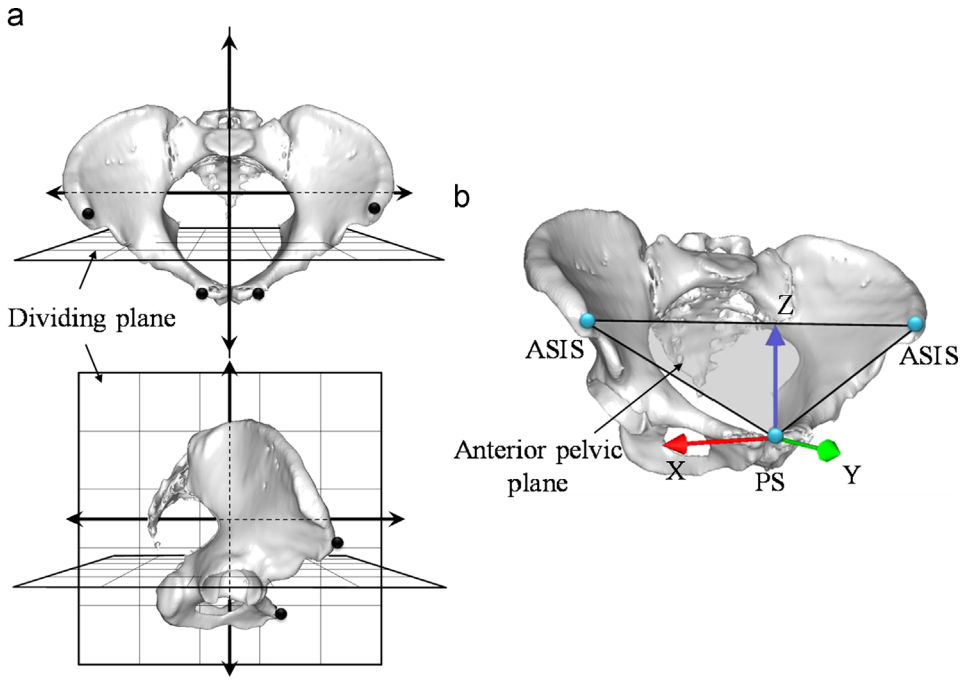


Fig. 1. (a) Dividing planes and the four most anterior points shown in coronal and sagittal views, based on the temporary coordinate system. (b) Pelvic ACS, *X*: the line parallel to a line connecting both ASISs, pointing to the right, *Y*: the line perpendicular to the anterior pelvic plane, pointing anteriorly, *Z*: the line perpendicular to both *X* and *Y*, pointing cranially, the origin coincides with PS.

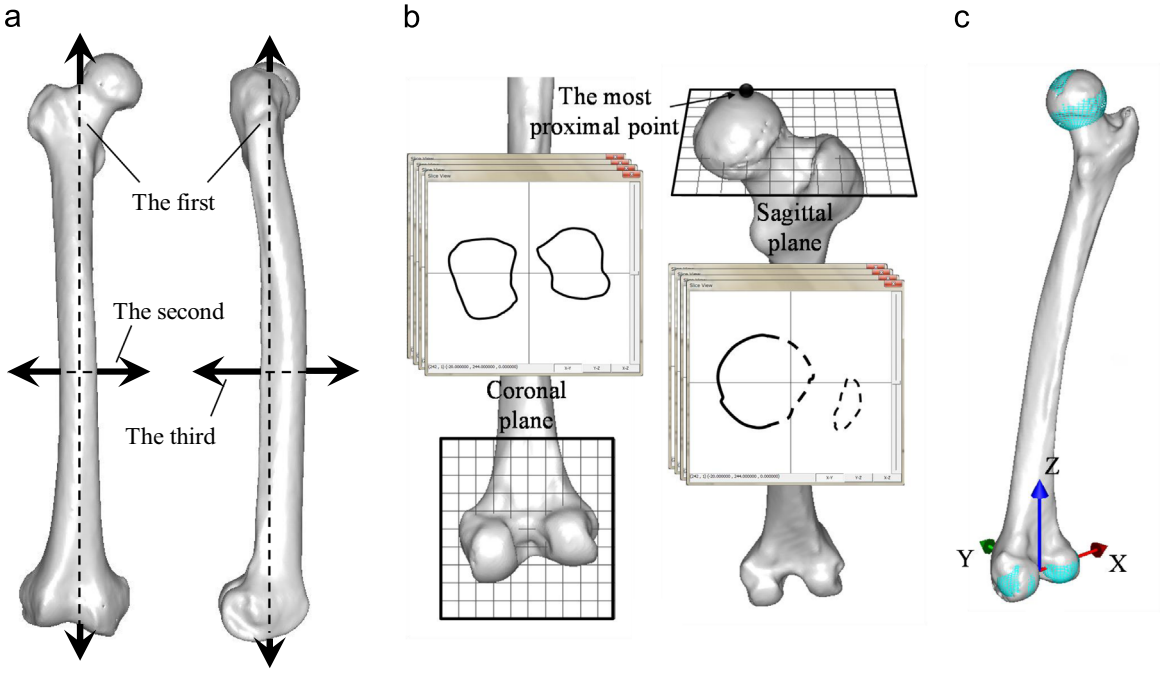


Fig. 2. (a) Temporary coordinate system constructed using principal component analysis of the femur. (b) Contour tracking to obtain the outlines of both posterior condyles and femoral head, based on the temporary coordinate system. (c) Femoral ACS, *X*: the line connecting the centers of both posterior condyles, pointing to the right, *Y*: the line perpendicular to the plane defined by the centers of both posterior condyles and the center of the femoral head, pointing anteriorly, *Z*: the line perpendicular to both *X* and *Y*, pointing cranially, the origin coincides with the midpoint of the centers of two both posterior condyles. Mesh spheres represented the approximated spheres of both posterior condyles and femoral head.

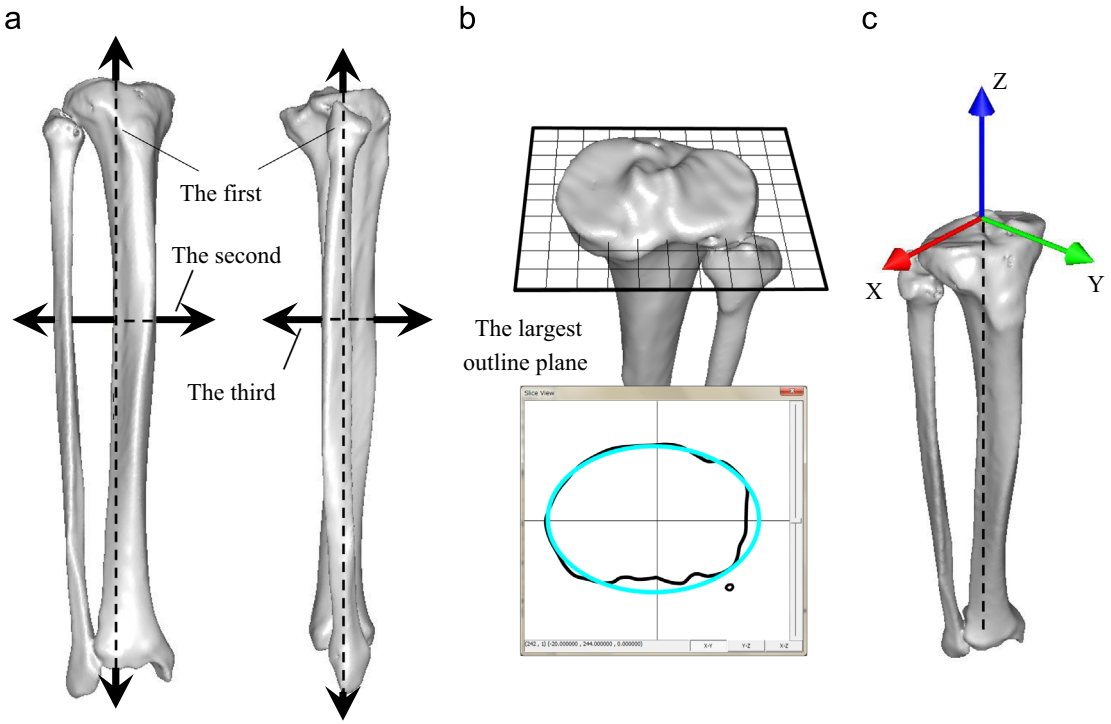


Fig. 3. (a) Temporary coordinate system constructed using principal component analysis of the tibia. (b) Contour tracking to obtain the largest outline, based on the temporary coordinate system. (c) Tibial ACS, *X*: the major axis of the approximated ellipse, pointing to the right, *Y*: the minor axis of the approximated ellipse, pointing anteriorly, *Z*: the line coinciding with the first principal component vector, pointing cranially, the origin coincides with the center of ellipse.

than those with automatic ACS (range, 9.6–18.8°) (Table 1). For the femur, since the length of the line between two epicondyles was short, consequently, the differences of its location caused larger errors in valgus–varus and rotation angles than the error in flexion–extension angles (Table 2). For the tibia, significant differences were found in the direction of the anteroposterior axis; the anteroposterior axes recommended by ISB were more external than those in the automatic ACS (range, 17.5–25.0°) (Table 3).

Table 1
Mean ± SD of rotational components, ICC(2,1) at each rotational component and its 95% confidence limit (upper-lower bounds) for comparison between the automatic ACS and ISB recommended pelvic ACS by five raters.

	Pelvis 1	Pelvis 2	Pelvis 3	Pelvis 4	Pelvis 5	ICC	95% CL
Inclination	9.6 ± 2.3 Anterior	16.4 ± 1.0 Anterior	18.8 ± 0.2 Anterior	11.1 ± 0.8 Anterior	12.7 ± 0.7 Anterior	0.91	0.98–0.50
Abduction–adduction	0.1 ± 0.2 Adduction	0.3 ± 0.5 Abduction	0.1 ± 0.3 Adduction	0.2 ± 0.6 Abduction	0.0 ± 0.4 Abduction	0.01	0.34–0.19
Rotation	0.0 ± 0.1 External	0.1 ± 0.2 Internal	0.1 ± 0.2 External	0.2 ± 0.2 Internal	0.0 ± 0.2 Internal	0.05	0.40–0.19

Table 2
Mean ± SD of rotational components, ICC(2,1) at each rotational component and its 95% confidence limit (upper-lower bounds) for comparison between the automatic ACS and ISB recommended femoral ACS by five raters.

	Femur 1	Femur 2	Femur 3	Femur 4	Femur 5	ICC	95% CL
Flexion–extension	0.9 ± 0.5 Flexion	1.1 ± 0.2 Flexion	0.9 ± 0.1 Flexion	0.9 ± 0.4 Flexion	1.1 ± 0.2 Flexion	0.33	0.69–0.14
Valgus–varus	1.5 ± 0.1 Varus	4.9 ± 0.1 Valgus	0.8 ± 0.0 Varus	3.7 ± 0.1 Valgus	3.2 ± 0.0 Valgus	0.99	1.00–0.99
Rotation	3.7 ± 4.0 External	4.3 ± 4.1 External	3.2 ± 1.5 Internal	2.3 ± 3.9 External	1.9 ± 2.6 External	0.13	0.479–0.02

Table 3
Mean ± SD of rotational components, ICC(2,1) at each rotational component and its 95% confidence limit (upper-lower bounds) for comparison between the automatic ACS and ISB recommended tibial ACS by five raters.

	Tibia 1	Tibia 2	Tibia 3	Tibia 4	Tibia 5	ICC	95% CL
Inclination	3.7 ± 0.9 Posterior	6.1 ± 2.2 Posterior	4.1 ± 2.4 Posterior	3.4 ± 1.8 Posterior	4.0 ± 3.7 Posterior	0.07	0.37–0.12
Valgus–varus	6.3 ± 1.5 Valgus	10.9 ± 1.5 Valgus	12.7 ± 1.4 Valgus	8.5 ± 2.2 Valgus	6.7 ± 4.3 Valgus	0.53	0.82–0.01
Rotation	20.9 ± 3.9 External	25.0 ± 6.2 External	17.5 ± 4.2 External	19.2 ± 5.9 External	21.0 ± 8.2 External	0.08	0.40–0.13

ICC(2,1) at each rotational component is also shown in the tables. Overall, these small values indicated that manual digitization by different raters would cause variation in constructing ACS even on the same 3D bone model.

4. Discussion

Since ACS constructed on a 3D bone model is considered as a criterion for all calculations, the most important quality in the construction of ACS is considered to be reproducibility, i.e., the ability to eliminate the influence of rater's bias on reference landmark selection. Moreover, anatomical landmarks should still be used to account for patient-specific bony geometry.

Automated methods for constructing a patient-specific coordinate system for 3D bone models of the lower extremities were developed in this study. However, in addition to the aforementioned ISB-recommended coordinate system, various other coordinate systems are needed to perform various evaluations, and it is difficult to determine which axis or landmark is appropriate. For the pelvis, the APP has almost become the gold standard as a reference plane for the alignment of the acetabular cup. Therefore, errors introduced during APP determination would substantially affect the final cup alignment. Results of a simulation indicated that cup anteversion and inclination errors could be as large as $\pm 10^{\circ}$ and $\pm 6^{\circ}$, respectively, assuming a 4-mm error in landmark localization (Wolf et al., 2005). In addition, pelvic tilt causes inaccuracies in APP determination (Lembeck et al., 2005). To reduce the influence of the aforementioned errors, a reliable coordinate system was developed in this study. The results indicated that ISB-recommended ACS had a greater anterior inclination compared with automatic ACS. For the femur, two flexion–extension axes are widely used to describe knee joint motion. One is the GCA in this study; another is the transepicondylar axis (TEA) (Olcott and Scott, 1999; Yoshino et al., 2001; Li et al., 2001). A previous study examined the sensitivity of kinematic data of the two axes, and found that rotation calculated using TEA was significantly different from that obtained using GCA (Most et al., 2004). Although Yoshioka et al. suggested that femoral epicondyles had been shown to be reliable anatomical landmarks (1987), it is difficult to manually localize the peaks of the epicondyles as a certain point with high reproducibility. For the tibia, the longitudinal axis is generally determined by digitizing superficial landmarks on both the proximal and distal articular surface (Sato et al., 2004; Yoshioka et al., 1989, Wu et al., 2002, Akagi et al., 2004). These landmarks are certainly bone-based anatomical landmarks; however, they are not easy to precisely determine, especially on a deformed articular surface. In contrast to these issues, our procedure has better reliability and the advantage of not being influenced by deformations. Our results indicated that the anteroposterior axis of ISB-recommended ACS had greater external rotation compared with automatic ACS, because the mediolateral axis was defined as the line connecting the medial and lateral malleoli, and tibial torsion resulted in greater external rotation. A clinical study reported significant differences in calculated extension and abduction moments caused by the effect of tibial torsion (Manal et al., 2002). Hence, we suggest that when applying a 3D bone model to kinematic study, care must be taken to clarify the coordinate system, the anatomical landmarks, and the relationship between them; thus, various landmarks such as mechanical axis, Akagi's line, etc. should be selected based on a coordinate system using common criteria, so that different studies can be compared or referred to. We believe that our automated coordinate system can be used as the common coordinate system to simplify the application and interpretation of biomechanical findings.

Miranda et al. also proposed an automated patient-specific ACS for the distal femur and proximal tibia (Miranda et al., 2010). But in order to obtain the true alignment of lower extremity, the proximal femur and distal tibia also have various important landmarks, 3D entire bone models should be acquired as much as possible. In any case, the most significant concern is the extra radiation exposure. However, although the effective radiation dose of 6 mSv from CT is about 10 times larger than that required for radiography, this dose is comparable to the annual effective dose from natural background radiation of about 3 mSv (Mettler et al.,

2008). In order to reduce the risk of radiation exposure, using different CT scan protocol would be effective. For example, thin slices are necessary only for the location around knee joint, therefore, thicker slices can be used for locations of other parts. Practically, only two slices around femoral head are sufficient to determine an approximated sphere in our method. Another limitation is the effort to construct 3D bone models. Although the automated methods for constructing patient-specific ACS offered high reproducibility and time efficiency for a given bone model, the reproducibility of the bone model construction is still required. However, automatic segmentation has already been possible except severe deformed joint, combining with the automatic segmentation would be the next step of this study.

Conflict of interest statement

None.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.jbiomech.2013.12.013.

References

Asano, T., Akagi, M., Tanaka, K., et al., 2001. In vivo three-dimensional knee kinematics using a biplanar image-matching technique. Clin. Orthopaed. Relat. Res. 388, 157–166.
Akagi, M., Oh, M., Nonaka, T., et al., 2004. An anteroposterior axis of the tibia for total knee arthroplasty. Clin. Orthopaed. Relat. Res. 420, 213–219.
Bergmann, G., Deuretzbacher, G., Heller, M., et al., 2001. Hip contact forces and gait patterns from routine activities. J. Biomech. 34, 859–871.
Della Croce, U., Cappozzo, A., Karrigann, D.C., et al., 1997. Bone position and orientation errors: pelvis and lower limb anatomical landmarks identification reliability. Gait Posture 5 (2), 156–157.
Della Croce, U., Cappozzo, A., Karrigann, D.C., 1999. Pelvis and lower limb anatomical landmarks calibration precision and its propagation to bone geometry and joint angles. Med. Biol. Eng. Comput. 37, 155–161.
Della Croce, U., Leardini, A., Chiari, L., et al., 2005. Human movement analysis using stereophotogrammetry Part 4: assessment of anatomical landmark misplacement and its effects on joint kinematics. Gait Posture 21, 226–237.
Eckhoff, D.G., Dwyer, T.F., Bach, J.M., et al., 2001. Three-dimensional morphology of the distal part of the femur viewed in virtual reality. J. Bone Joint Surg. (Am.) 83-A, 43–50. (Suppl 2 (Pt 1).
Enomoto, H., Nakamura, T., Waseda, A., et al., 2013. A novel and reproducible reference axis for distal tibial axial rotation. J. Arthroplasty 28, 788–791.
Gonzalez-Ochoa, C., McCammon, S., Peter, J., 1998. Computing moments of objects enclosed by piecewise polynomial surfaces. ACM Trans. Graphics 17 (3), 143–157.
Grood, E.S., Suntay, W.J., 1983. A joint coordinate system for the clinical description of three-dimensional motions: application to the knee. J. Biomech. Eng. 105, 136–144.
Kang, Y., Engelke, K., Fuchs, C., et al., 2005. An anatomical coordinate system of the femoral neck for highly reproducible BMD measurements using 3D QCT. Comput. Med. Imag. Graphics 29, 533–541.
Lembeck, B., Mueller, O., Reize, P., et al., 2005. Pelvic tilt makes acetabular cup navigation inacurate. Acta Orthopaed. 76, 517–523.
Lewinnek, G.E., Lewis, J.L., Tarr, R., 1978. Dislocations after total hip-replacement arthroplasties. J. Bone Joint Surg. (Am.) 60, 217–220.
Li, G., Lopez, O., Rubash, H., 2001. Variability of a three-dimensional finite element model constructed using magnetic resonance images of a knee for joint contact analysis. J. Biomech. Eng. 123, 341–346.
Manal, K., McClay, I., Richards, J., et al., 2002. Knee moment profiles during walking: errors due to soft tissue movement of the shank and the influence of the reference coordinate system. Gait Posture 15, 10–17.
Mettler, F.A., Huda, W., Yoshizumi, T.T., et al., 2008. Effective doses in radiology and diagnostic nuclear medicine: a catalog. Radiology 248 (1), 254–263.
Miki, H., Yamanashi, W., Nishii, T., et al., 2007. Anatomical hip range of motion after implantation during total hip arthroplasty as measured by a navigation system. J. Arthroplasty 22, 946–952.
Miranda, D.L., Rainbow, M.J., Leventhal, E.L., et al., 2010. Automatic determination of anatomical coordinate systems for three-dimensional bone models of the isolated human knee. J. Biomech. 43, 1623–1626.
Most, E., Axe, J., Rubash, H., Li, G., 2004. Sensitivity of the knee joint kinematics calculation to selection of flexion axes. J. Biomech. 37, 1743–1748.
Olcott, C.W., Scott, R.D., 1999. The ranawat award. Femoral component rotation during total knee arthroplasty. Clin. Orthopaed. Relat. Res. 367, 39–42.
Pearson, K., 1901. On lines and planes of closest fit to systems of points in space. Philos. Mag. 2, 559–572.
Sato, T., Koga, Y., Omori, G., et al., 2004. Three-dimensional lower extremity alignment assessment system. Application to evaluation of component position after total knee arthroplasty. J. Arthroplasty 19, 620–628.
Tashman, S, Anderst, W., 2003. In-vivo measurement of dynamic joint motion using high speed biplane radiography and CT: application to canine ACL deficiency. Trans. ASME 125, 238–245.
Walker, P.S., Kurosawa, H., Rovick, J.S., et al., 1985. External knee joint design based on normal motion. J. Rehab. Res. Dev. 22, 9–22.
Wolf, A., Digioia III, A.M., Mor, A.B., 2005. Cup alignment error model for total hip arthroplasty. Clin. Orthopaed. Relat. Res. 437, 132–137.
Wu, G., Siegler, S., Allard, P., et al., 2002. ISB recommendations on definitions of joint coordinate system of various joints for the reporting of human joint motion-part I: ankle, hip, and spine. J. Biomech. 35, 543–548.
Yoshino, N., Takai, S., Ohtsuki, Y., et al., 2001. Computed tomography measurement of the surgical and clinical transepicondylar axis of the distal femur in osteoarthritis knees. J. Arthroplasty 16, 493–497.
Yoshioka, Y., Siu, D.W., Cooke, T.D.V., 1987. The anatomy and functional axes of the femur. J. Bone Joint Surg. (Am.) 69-A, 873–880.
Yoshioka, Y., Siu, D.W., Scudamore, R.A., et al., 1989. Tibial anatomy and functional axes. J. Orthopaed. Res. 7, 132–137.